

Skills Summary

Polynomial Zeros and Turning-Point Constraints

Use this reference while completing your worksheet or when studying.

1. Factor, then read the zeros

Rule. The real zeros of a polynomial are the x -values where every factor can be made zero: factor completely, set each linear factor to 0, and solve.

Order of attack: common factor first, then a recognisable pattern (difference of squares, quadratic in x^2), then the Rational Root Theorem — any rational zero of a monic polynomial divides the constant term.

Example: Find the real zeros of $f(x) = x^2 - x - 12$.

Step 1. Monic quadratic with no common factor, so look for a factor pair of the constant term rather than reaching for the quadratic formula.

Step 2. -4 and 3 multiply to -12 and add to -1 , so $f(x) = (x - 4)(x + 3)$.

$$\Rightarrow x = 4, -3$$

Try it: Find the real zeros of $x^2 + 2x - 15$.

check: $x = 3, -5$

2. Multiplicity: cross or touch

Rule. The exponent on a linear factor is that zero's **multiplicity**.

Odd multiplicity: the factor changes sign, so the graph *crosses* the axis. **Even** multiplicity: no sign change, so the graph *touches* and turns back — and that touch point is itself a turning point.

Multiplicities add up to the degree, so they also tell you how many crossings to expect.

Example: Describe the zeros of $f(x) = (x + 1)^2(x - 4)$.

Step 1. The form is already factored, so nothing needs solving — read each zero and its exponent straight off.

Step 2. $x = -1$ comes from a squared factor (multiplicity 2, even, so the graph touches); $x = 4$ comes from a first-power factor (multiplicity 1, odd, so the graph crosses).

$$\Rightarrow x = -1, 4$$

Try it: Find the zeros of $x^2(x - 5)$ and say which one is a touch.

check: touch at 0, cross at 5

3. What the degree allows

Bounds. A polynomial of degree n has **at most** n real zeros and **at most** $n - 1$ turning points.

Read them both ways. Forwards: degree 6 allows up to 5 turns. Backwards: a graph showing m turning points needs degree at least $m + 1$.

And a lower bound: the graph must turn between two consecutive distinct real zeros, so k distinct zeros force at least $k - 1$ turns.

Example: What is the greatest number of turning points a degree-6 polynomial can have?

Step 1. This is the forwards reading of the bound, so subtract one from the degree — no factoring is involved.

Step 2. $n - 1$ with $n = 6$.

$\Rightarrow$ 5 turning points.

Try it: A graph has 3 turning points. What is the smallest degree it could have?

check: 4

4. End behaviour of a rational function

Rule. Compare degrees of numerator and denominator.

Numerator degree *smaller*: the limit is 0, so $y = 0$. Degrees *equal*: the limit is the ratio of leading coefficients. Numerator degree *larger*: no horizontal asymptote.

The move that proves it: divide every term by the highest power of x ; each leftover fraction goes to 0.

Example: Find $\lim_{x \rightarrow \infty} \frac{2x^2 - x}{x^2 + 3}$.

Step 1. Top and bottom have the same degree, so expect a nonzero horizontal asymptote — divide through by x^2 to see it.

Step 2. The expression becomes $\frac{2 - 1/x}{1 + 3/x^2}$, and both small fractions go to 0.

$\Rightarrow$ 2

Try it: Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 1}{2x^3 - x}$.

check: 5/2